JASWIN CHINTHALA

Cape Town, South Africa (SA Permanent Resident) | jaswinchinthala08@gmail.com | +27 67 931 1089
linkedin.com/in/jaswin-chinthala | github.com/jaswin-codes | jaswin.me

First-year Mechatronics student seeking vacation work and research roles in EV/power systems and AI safety.

EDUCATION

University of Cape Town – BSc Eng Electrical Engineering (Mechatronics) Feb 2026 – Nov 2029 (expected)
- Distinctions (Semester 1, 2026): Engineering Calculus, Introduction to Electrical Engineering, Computer Science I (Python).

Blouberg International School – Cambridge A-Levels: Mathematics (A), Physics (B) 2019 – 2025

AWARDS

National Chess Champion · Best F1 Score, StatsGeeks Data Science Hackathon (2026) · Student of the Year, Grade 11 · President's Award (Bronze)

EXPERIENCE

UCT Racing – Formula Student Team Feb 2026 – Present
EV Research Subteam Lead (Aug 2026 – Present); Power Subsystem Member (Feb – Jul 2026)
- Set up and lead the team's new EV research subteam, building its research roadmap (rules compliance, benchmarking winning EV teams, cost model) for a design-only Concept Class entry at Formula Student UK 2027.
- Co-wrote the cooling-system documentation (radiator fan integration) and a report on real-time vehicle telemetry and battery thermal management.

UCT Electric Mobility Lab (EMi Lab) – Tukzie electric tuk-tuk project Jun – Jul 2026
Vacation Work Student – Electric Drive Reliability & Power Electronics
- Modelled battery management (cell balancing, state-of-health tracking) in MATLAB, Simulink and Simscape in a 5-person team.
- Compared 4 swappable-battery infrastructure designs for electric tuk-tuks and recommended 1 in a written technical report.

UCT D-School Makerspace Oct 2026 – Present
Studio Assistant (part-time)
- Help run student workshops and microelectronics projects for beginners.

Scientific Caribbean Foundation & UCT May 2026
Student Researcher – 2-week Research Methodology Programme
- Built a Python simulation adapting the University of Florida's FreshID produce-freshness method to low-cost sensing; red-edge (760 nm) and near-infrared (1100 nm) bands gave the earliest freshness signal in simulation. Presented a poster to an academic panel.

PROJECTS

Water Level & Loadshedding Alarm – Electrical Engineering II design project Oct 2026
- Designed an analogue alarm (no microcontroller) using LM324 comparators, CMOS logic and two 555 timers; met all 6 required warning states in LTspice, then built and tested it on a breadboard.
- Wrote the technical report (threshold and timing calculations, simulation vs. bench results) and traced 3 wiring faults by comparing measured node voltages with the simulation.

AfriGuard – AI Safety Red-Teaming (team of 4, Global South AI Safety Hackathon) Jun 2026
github.com/ubayd-hattas/AfriGuard
- Ran 1,120 jailbreak tests on 4 frontier LLMs in 7 South African languages: attacks succeeded 50.1% of the time, more than double the English rate (24.4%). Co-built the test pipeline and a Streamlit results dashboard.

Digital Minds Research Sprint – Apart Research (team of 2) Aug 2026
github.com/UbaJaz/Digital_Minds_Research
- Tested whether LLMs predict their own behaviour better than outside observers: a simple 21-feature text classifier matched or beat the models' own self-predictions in 6 of 10 conditions.
- Built the Python experiment pipeline: OpenRouter API client, generation and prediction runners, checkpointed runs.

StatsGeeks Data Science Hackathon (team of 3) Sep 2026
github.com/jaswin-codes/StatsGeeks
- Placed 1st of 6 teams for Best F1 Score (macro-F1 0.75) with a location-aware Random Forest that classified building age from satellite pixels in a new city (Madrid → Amsterdam) from few labelled examples.

SANReN Cyber Security Challenge – Capture the Flag (team lead, team of 4) Aug 2026
- Led a team of first-time CTF players through the qualifying round: solved 34 challenges for 3,000+ points across digital forensics, cryptography, web exploitation and binary analysis.

LEADERSHIP

Treasurer, IEEE UCT Student Branch – manage the budget and records; plan spend on workshops and events. Oct 2026 – Present
Co-Founder & Secretary, UCT Maker Society – cross-faculty society for student builders. Oct 2026 – Present
Class Representative, Intro to Electrical Engineering – 1 of 6 elected reps for 170 students. Feb – Jun 2026

SKILLS

Technical: Python, MATLAB/Simulink/Simscape, LTspice, Java (Computer Science II, in progress), Git/GitHub
Electronics & making: op-amp comparators, CMOS logic, 555 timers, breadboard prototyping and debugging, 3D printing
AI/ML: LLM evaluation and red-teaming, scikit-learn, Streamlit
Languages: English, Telugu